

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – CONCEPT MAP

|                  |                                                                                                                       |               |                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------|
| Course Code:     | ITA0312                                                                                                               | Course Title: | Mobile computing                |
| CO Assessed:     | CO1                                                                                                                   | Assessment:   | Assessment Tool 1 – Concept Map |
| Weightage:       | 60 % of CIA                                                                                                           | Total Marks:  | 20 marks                        |
| CO1 Statement:   | Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems. |               |                                 |
| Student Name:    | P.Ranjithkumar                                                                                                        | Reg. No.:     | 192411119                       |
| Section / Batch: | 2024                                                                                                                  | Date:         | 04.10.2026                      |

### Code of Conduct

*I P.Ranjithkumar (192411119)\_\_\_\_\_ (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the Student: P.Ranjithkumar\_\_\_\_\_*

### Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

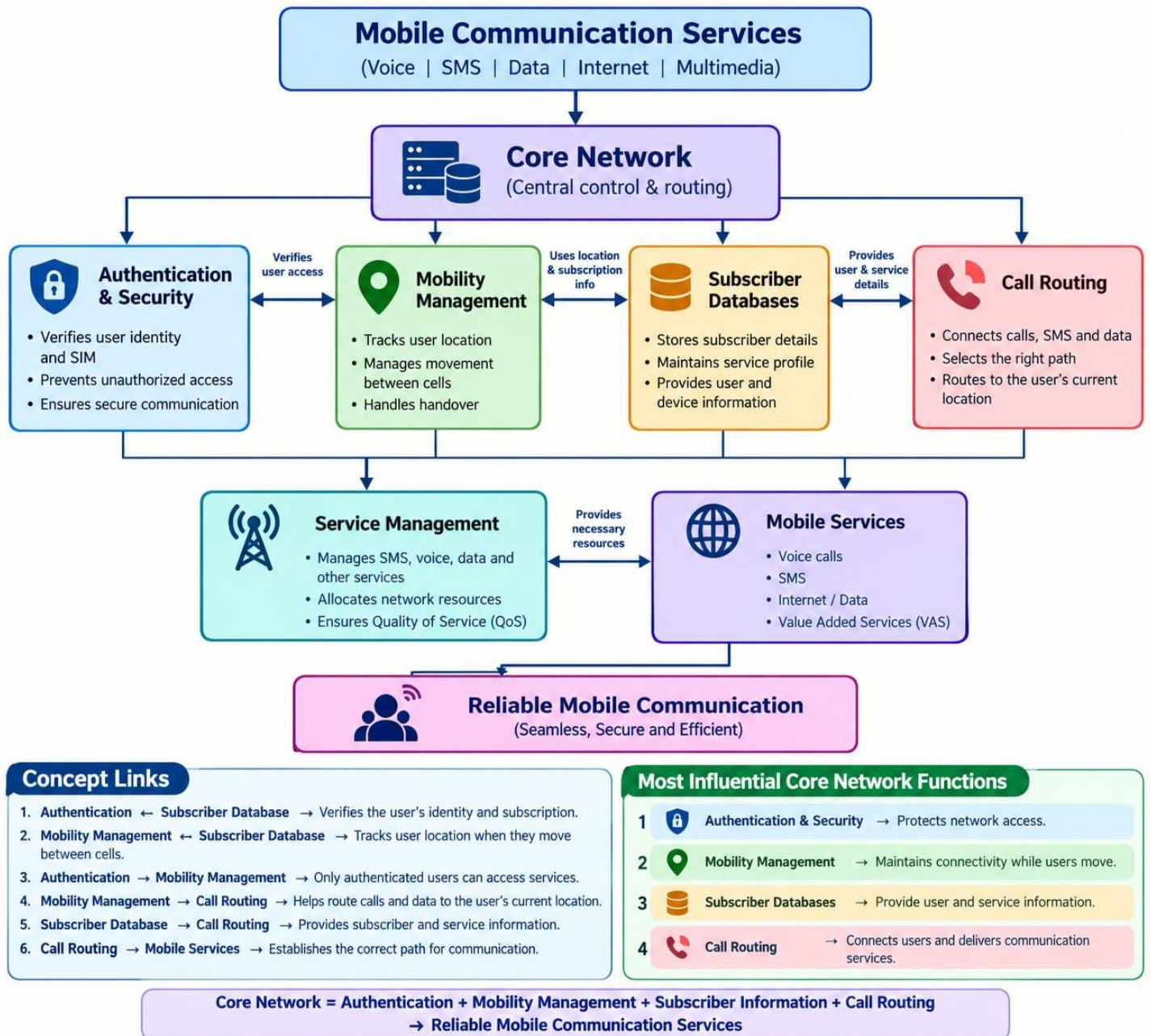
## Annexure A - QUESTIONS

- Design a concept map that evaluates the role of the core network in delivering mobile communication services.

Relate authentication, mobility management, subscriber databases, and call routing.

Determine the most influential core network functions.

Justify your evaluation through concept linkages.



## Justification of the Concept Map

Authentication & Security – Verifies the user's identity and prevents unauthorized access to mobile services.

Subscriber Database – Stores important information such as user identity, subscription details, and service information.

Mobility Management – Tracks the user's location and maintains connectivity when the user moves between different cells.

Call Routing – Finds the correct path and connects calls, SMS, and data to the intended user.

Service Management – Manages voice, SMS, internet, and other mobile services while supporting Quality of Service (QoS).

Concept Link: Authentication → Mobility Management – A user must be authenticated before accessing network services while moving between cells.

Concept Link: Subscriber Database → Call Routing – Subscriber information helps the network identify the user and route communication correctly.

Concept Link: Mobility Management → Call Routing – The user's current location helps the network route calls and data to the correct location.

Overall Evaluation – The Core Network is essential because authentication, subscriber information, mobility management, and call routing work together to provide secure, continuous, and reliable mobile communication services.

| Criteria                         | Weightage | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    |
|----------------------------------|-----------|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20        | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts          | 30        | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |
| Problem-Solving Approach         | 20        | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |
| Accuracy of Solution             | 20        | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |
| Clarity                          | 10        | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |